# ISM Note 1

Yang Chen

3/5/2026

# 1. The interstellar medium (ISM)

The true name of this course should be ***The physics of the diffuse baryons***, which is more precise to describe the content of this course. The "baryons" here refer to the normal matter in our universe, which is typically made up of protons, neutrons, and electrons. Beisdes, over 99.99% of the baryons in the universe are in the form of **plasma**, so this means we are dealing with ionized matters in most cases, only in special situations will we consider the neutral ones. The "diffuse" here means that the density of these baryons is very low, to make this concept more concrete, we consider the non-diffuse baryons, which are those in compact objects like stars. So for our researching here, we will not consider those compact objects and mainly care about the diffuse "gases".

**The mean density of the ISM is about 1 particle per cubic centimeter (1 cm$^{-3}$).** In other areas of physics, this is very dilute and can be approximated as the vacuum, but in this ISM course this is not the case. Though the density is really low, there are active interactions between the ISM and stars: the ISM can concentrate to form stars, and the stars can also inject energy and mass into the ISM through stellar winds and supernova explosions.

After knowing the "diffusive" and "non-diffusive" objects and their interactions, a naive question one can ask is: what is the ratio of the total mass of the diffuse baryons to the total mass of the stars? Roughly it is about 1:1, this might have something to do with the feedback mechanisms between the ISM and stars. Up to now we have give the raw physical picture for the ISM.

# The big picture of our universe

Initially the universe is mainly made of dark matter uniformly distributed. And then the quantum fluctuations acted as **perturbations** to the dark matter, which will cause the dark matter to shrink and grow by virtue of a positive feedback, which is the so-called **gravitational instabilities**. Then it will form the large scale structure of the universe, which is also called t**he cosmic web**, and its characteristic scale is about $10^2$ **Mpc**. The cosmic web is like the skeleton of the universe, and the gas will flow through the filaments and fall into the dark matter halos motivated by the gravity. For astrophysicists, though we cannot observe the dark matter directly, we can observe the number densities of galaxies and stars to infer the distribution of dark matter. **The gas flow along the filaments are called intergalactic medium (IGM)**.

The cosmic web image is as follows (from the internet)

Fig 1. The cosmic web structure.

In the DM halos, there are thousands of galaxies in the galaxy clusters, and **the gas within the galaxy clusters is called the intracluster medium (ICM)**. Its characteristic scale is about $10^1$ **Mpc**. By the way, the galaxy clusters are the largest gravitational bounded structures in the universe, larger objects like the cosmic web structure are not gravitational bounded.

Then we have the galaxies, and **the gas around the galaxies is called the circumgalactic medium (CGM)**, and its characteristic scale is about $10^2$ **kpc**.

Finally, **we have the ISM, which is the gas within the galaxies**, and its characteristic scale is about $10^1$ **kpc**.

This course will cover all the gases we mentioned above, which constitute the diffuse baryons in the universe.

# 2. The galaxies

The galaxies can be classified into two categories, the spiral galaxies and the elliptical galaxies. They are also called normal galaxies. The first question is, what is the number density of the galaxies? Is it a trivial value in the whole universe or does it satisfy some special law? The answer is given by the **Schechter function**, which is also known as the luminosity function of the galaxies. The Schechter function is given by

$$\Phi(L) = \frac{dn}{dL} = \Phi_0 \left(\frac{L}{L_*}\right)^{-\alpha} e^{-L/L_*}$$

where $\phi_*$ is the number density of galaxies with milky way luminosity, $L_*$ is the milky way luminosity, and $\alpha$ is the faint-end slope. The Schechter function describes the number density of galaxies as a function of their luminosity. There is no specific luminosity that can characterize the distribution since it is a power law.

The sketch of the Schechter function is as follows (from the internet)

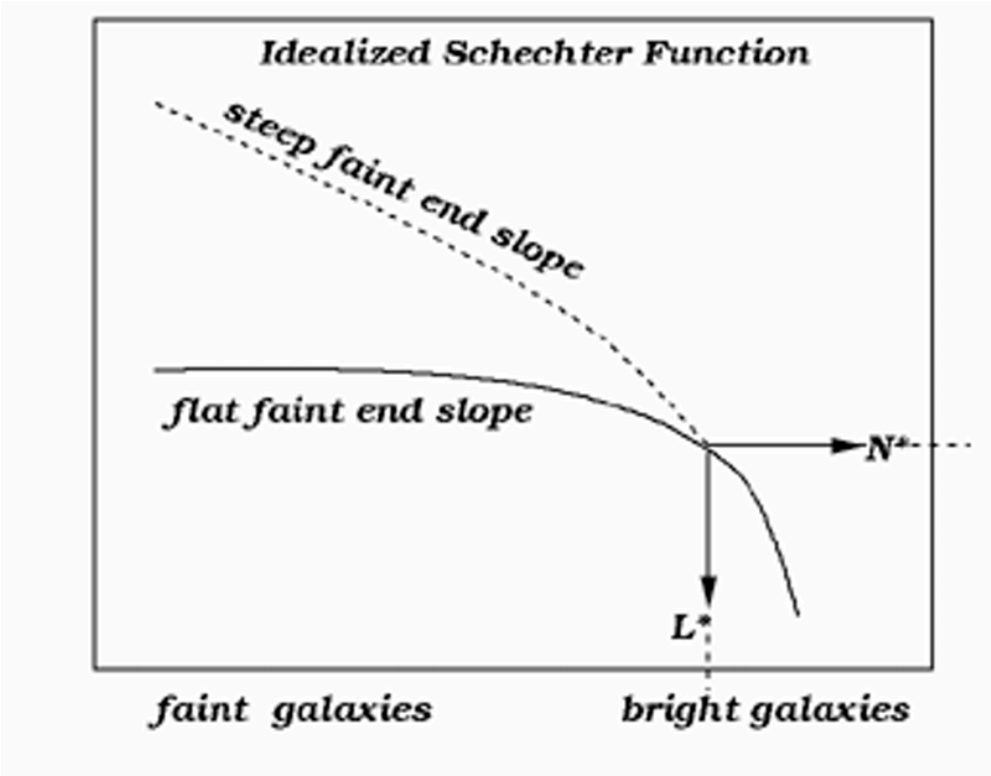


Fig 2. The Schechter function (Luminosity function).

From the image, we can see that the luminosity(x-axis) covered mainly 4 orders of magnitude, and for the region

$$L/L_* > 10^0$$

, there are normal galaxies, and since the elliptical galaxies are more massive while the spiral galaxies are less massive, the elliptical galaxies mainly occupy the right area of this regiion. And for the region

$$L/L_* < 10^0$$

, there are dwarf galaxies. The dwarfs are usully irregular shaped, because their mass are too small to power for the least free energy state for a "sphere".

We can get a theoretical prediction of the luminosity function from the dark matter theory, which is also shown in figure 2 above. But except for the turning point, the prediction overestimates the number density of galaxies. The reason for the "lower" real curve is mainly due to the feedback mechanism. But here we will not go into the details and leave it to the future lectures.

# Spiral / Disk galaxies

The spiral galaxies are very flat and we can abstract it as 2D disks for analytical purposes. The face-on view and edge-on view sketches are as follows

Fig 3. The face-on view and edge-on view of the spiral galaxy.

We usually use **surface density** to describe the properties of the spiral galaxies, and the surface density is defined as $\Sigma = \int \rho(r, \phi, z) dz$, where $\rho$ is the volume density and $z$ is the vertical direction. Usually for a spiral galaxy, the surface density satisfies $\Sigma(r) \propto e^{-r/H}$, where $H$ is the scale length.

So why the spiral galaxy form a disk-like structure? The reason is rotation. Horizontally, the galaxy is supported by rotation. Vertically, the galaxy has a height and not shrink into a line because of random motion -- thermal pressure balances with the gravity.

The rotational velocity roughly satisfies $v_{\mathrm{rot}}(r) \propto \frac{GM(r)}{r}$, and then we can plot the rotational curve, which is as follows

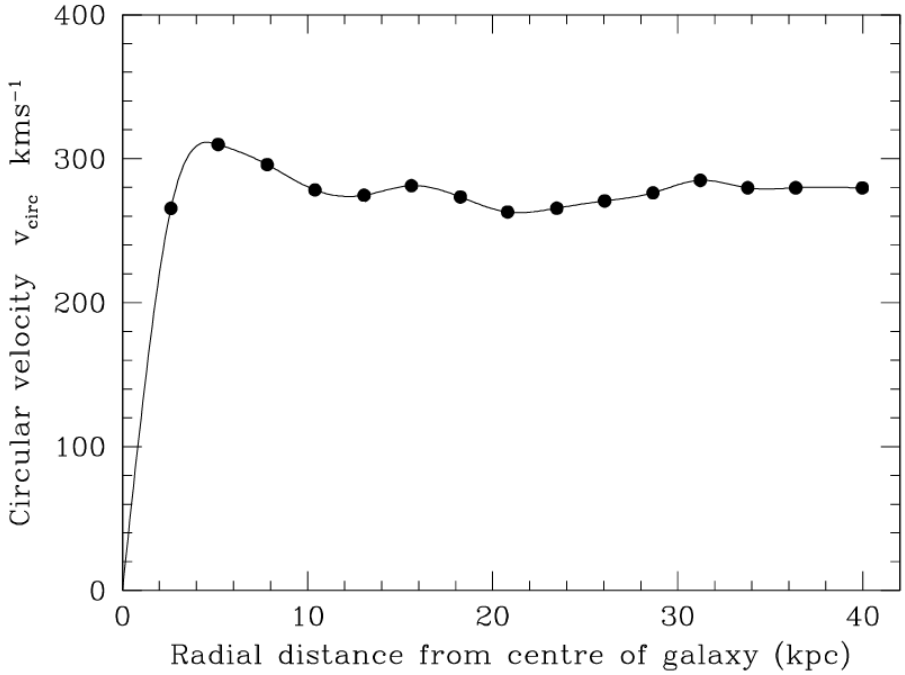


Fig 5. The rotational curve of the spiral galaxy.

The observational data is quite different from the theoretical prediction in the **large r region**, and the reason is that there are a large amount of dark matters in the outer region of the galaxy, which can provide extra gravity to support the rotation, so in this region the rotational velocity does not decrease but keep "flat". Knowing the numerous existence of dark matter, then intuitively we can define a quantity named **baryon fraction**,

$$f_b = \frac{M_{\mathrm{b}}}{M_{\mathrm{total}}}$$

The common value of the baryon fraction for dwarfs is about 1%-5%, while for normal galaxies it is about 5%-10%.

# Spiral arms

For the special shape of the spiral galaxies, another question one can ask is: why do spiral galaxies have spiral arms? This spiral arm is a kind of non-axisymmetric structure, which seems theoretically difficult to deal with, but fortunately we can use Fourier decomposition to analyze it. The Fourier decomposition is given by

$$f(r,z,\phi) = \sum_{m=-\infty}^{\infty} f(r,z)e^{im\phi}$$

where m is called the azimuthal number, and different m corresponds to different arm numbers.

The reason for the formation of the spiral arms is various, and a famous one is **the density wave theory**, another one is **the gravitational instability**. The core idea of the density wave theory is that the spiral arms are like traffic jams, which are not material arms, but the density waves. For the gravitational instability theory, the idea is that the spiral arms are formed due to the gravitational instability of the disk, which can be analyzed by **the Toomre's Q parameter**, defined by

$$Q = \frac{c_s \kappa}{\pi G \Sigma}$$

where $c_s$ is the sound speed, $\kappa$ is the epicyclic frequency, which is defined by $\kappa = \sqrt{\frac{2\Omega}{r}\frac{d\Omega}{dr}}, and$\Sigma$is the surface density. If$Q < 1$, the disk is gravitationally unstable, and if$Q > 1$, the disk is gravitationally stable, while if$Q \approx 1$, the disk is marginally stable.

# Analysis of the Toomre's Q parameter

## $c_s$ -- sound speed

The sound speed tells you **the transmission speed of the perturbation in pressure**, while it is also a speed of information in the fluid.
We can easily derive $P \propto \rho\gamma$, which means the thermal pressure is highly related to the sound speed. Thus intuitively the higher the sound speed, the stronger the thermal motion.

## $\kappa$ -- epicyclic frequency

The epicyclic frequency is the frequency of the small oscillation of a particle around its circular orbit. **It corresponds to the rotation physically.**

From the Toomre's Q parameter and our disscussion above, we can see that thermal pressure + rotation can stabilize the disk, while the self-gravity($G\Sigma$) can destabilize the disk.

If the disk is gravitationally unstable, it will fragment into clumps, and the clumps will collapse to enhance the star formation.